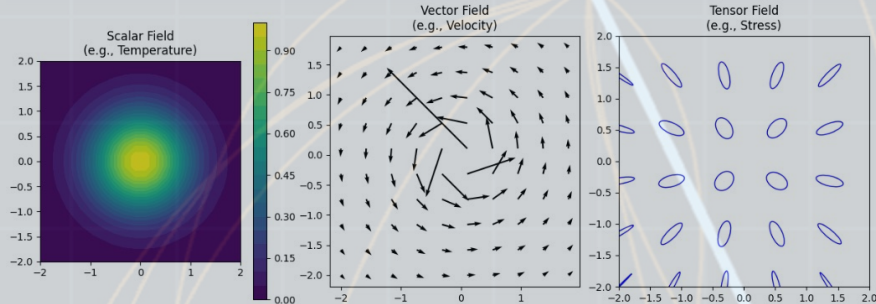




ENR-275 Classical Mechanics



Xiang Li
Fall 2026



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Course Syllabus and Discussion

In classical physics, if you know everything about a system at some instant of time, and you also know the equations that govern how the system changes, then you can predict the future.

- Leonard Susskind *Classical mechanics the theoretical minimum*

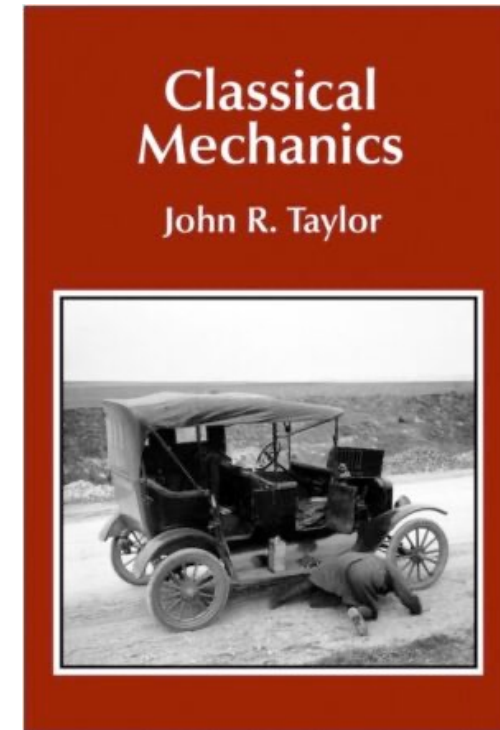
Course Syllabus and Discussion

Welcome to “Classic Mechs”:

- Pre-quantum
- Differential equations
- Linear algebra
- Little bit of iterative questions

Pre-requisition skills:

- ☐ Algebraic calculation and trigonometry
- ☐ Recall differential equations and vector calculus



What is classic mech to me:

What's next next next:

Modern "EVERYTHING"

Modern physics

Modern Robotics

AI

What's next next:

Continuum mechanics

Control systems

Data science and machine learning

What's next:

Complex analysis

Vector calculus

Fourier and Laplace transforms

PDE



Least Action Principle, Lagrangian function, and Hamilton's equations

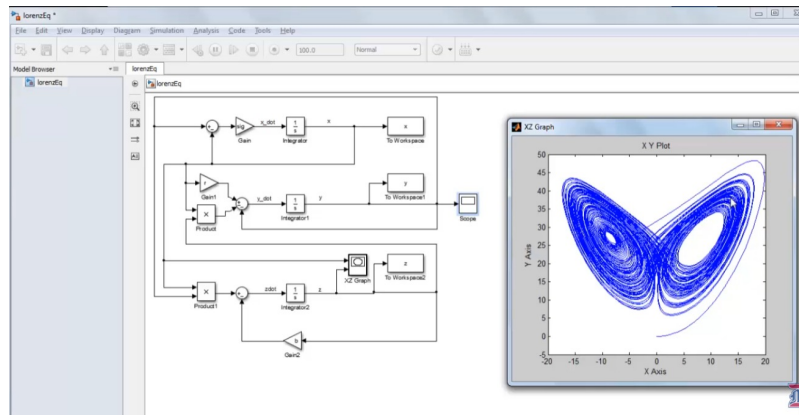
Calculus, ODE, Systems of ODE, Eigenvalues and Eigenvectors
Non-linear systems, Numerical and computational method



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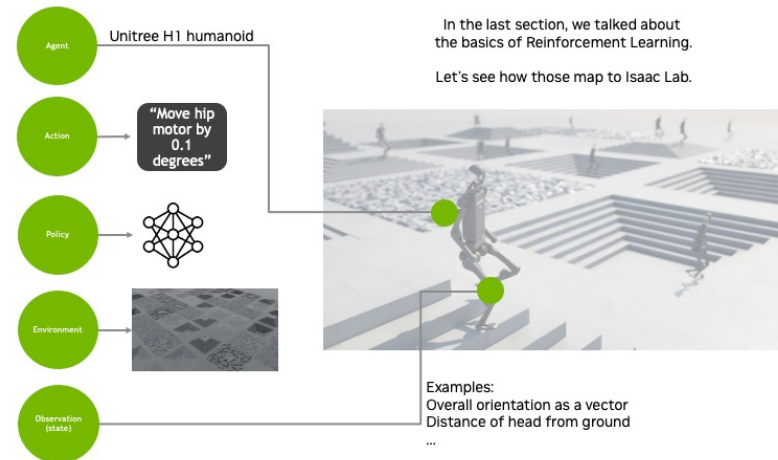
Example of “Modern”

Dynamic control of a chaotic system



From Rick Hill's Modern Dynamic System lecture channel, 2015

Reinforced learning for robotics

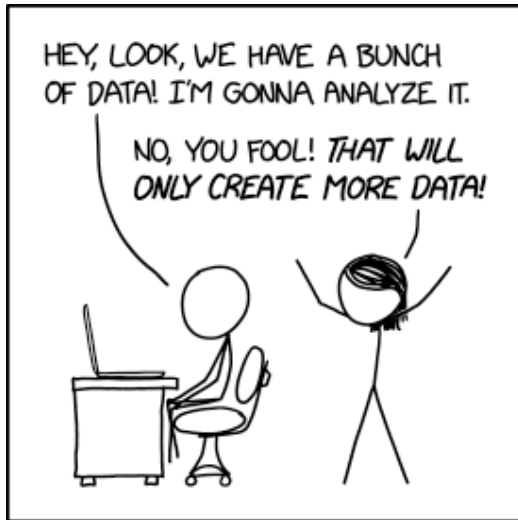


<https://docs.nvidia.com/learning/physical-ai/getting-started-with-isaac-lab/latest/train-your-first-robot-with-isaac-lab/01-what-is-reinforcement-learning.html>. 2020

Robot foundation model



<https://generalistai.com/blog/gen-1.5>, 2026



<https://xkcd.com/2582/>

Classic Mechs EP1: Analytical and probabilistic modeling

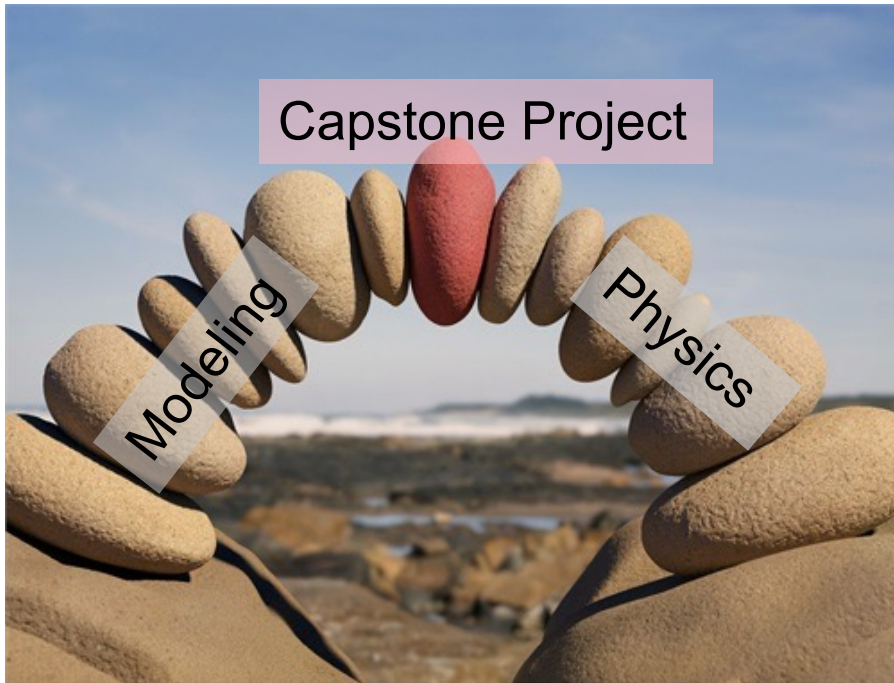
Classic Mechs EP1: Analytical and probabilistic modeling

“Pre-lab”:

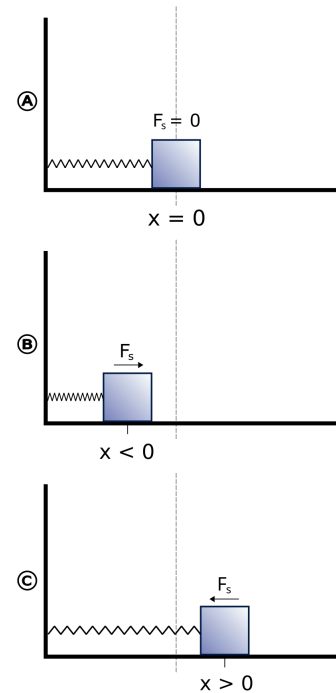
- Review Newton's law
- Review Cartesian coordinates
- Review equation of motion (figure 1.9 and 1.14), will be in the quiz on ~~Friday~~Next Wednesday?
- New Pre-lab tasks: read Taylor's chapter 1 and 2 and let me know if you have any trouble with the math by submit a polling (will be emailed to you guys shortly).

**Classic Mechs: you will do great in this class
if you can do the following two things:**

1) Make connections between...

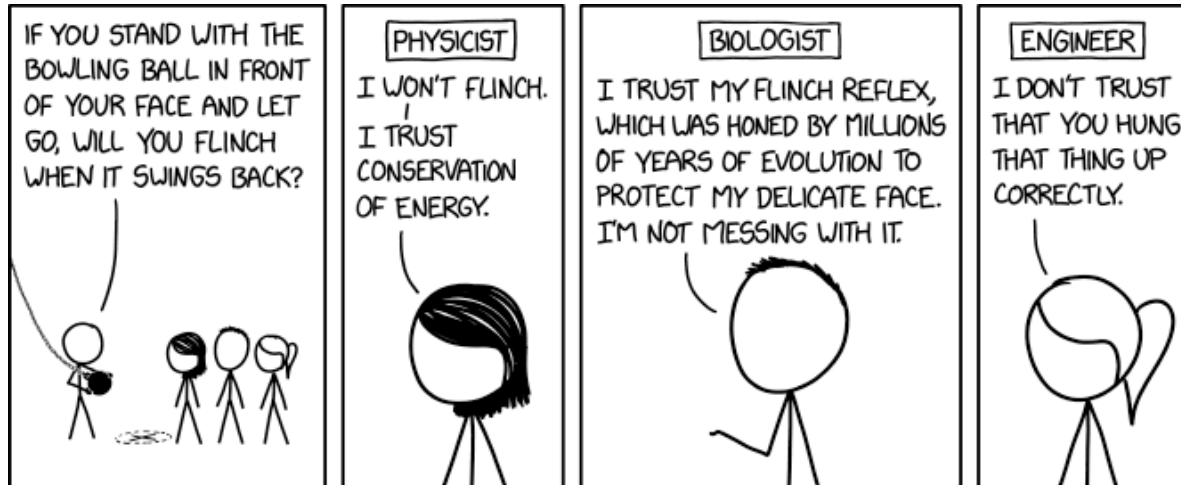


Whiteboard example: solving harmonic oscillator with ODE



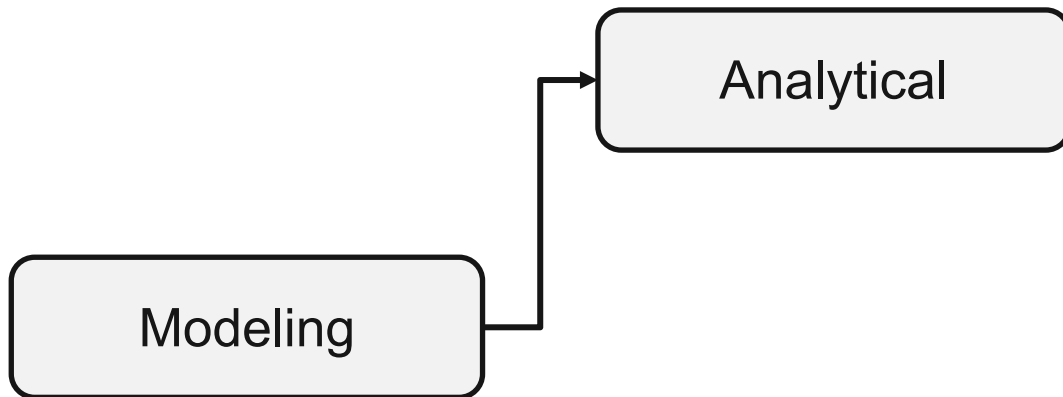
2) Generalize your model

Whiteboard example: derive conservation of energy from newton's law



<https://xkcd.com/2539/>

Analytical modeling (1/2)



First Principle

(1.10)

Power law

part of (4.2)

Chain rule

to get (1.47)



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Analytical modeling (2/2)

Section 1.7 Two-Dimensional Polar Coordinates

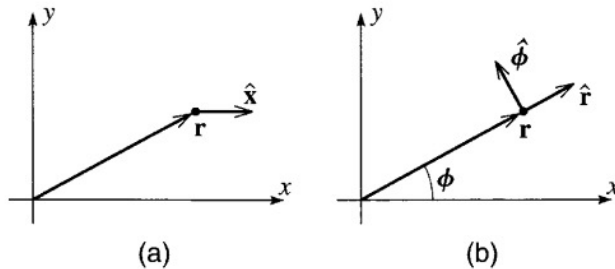


Figure 1.11 (a) The unit vector $\hat{x}$ points in the direction of increasing x with y fixed. (b) The unit vector $\hat{r}$ points in the direction of increasing r with ϕ fixed; $\hat{\phi}$ points in the direction of increasing ϕ with r fixed. Unlike $\hat{x}$, the vectors $\hat{r}$ and $\hat{\phi}$ change as the position vector $\mathbf{r}$ moves.

$$\begin{aligned}\Delta \hat{r} &\approx \Delta \phi \hat{\phi} \\ &\approx \dot{\phi} \Delta t \hat{\phi}.\end{aligned}\tag{1.41}$$

$$\frac{d\hat{r}}{dt} = \dot{\phi} \hat{\phi}.\tag{1.42}$$

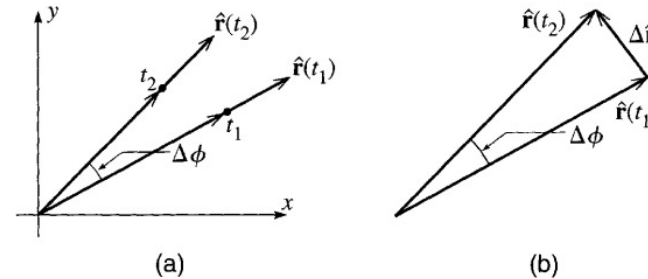


Figure 1.12 (a) The positions of a particle at two successive times, t_1 and t_2 . Unless the particle is moving exactly radially, the corresponding unit vectors $\hat{r}(t_1)$ and $\hat{r}(t_2)$ point in different directions. (b) The change $\Delta \hat{r}$ in $\hat{r}$ is given by the triangle shown.

$$\frac{d\hat{\phi}}{dt} = -\dot{\phi} \hat{r}.\tag{1.46}$$

$\hat{\phi}$ and $\vec{\phi}$ diff!

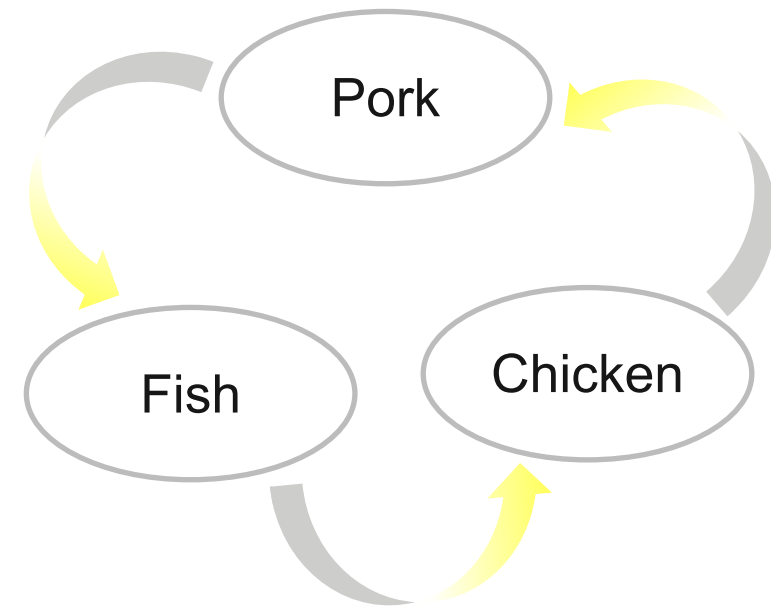


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Probabilistic modeling, an example

I noticed that Simple Servings at the U is doing a rotation of proteins: Pork, Fish and Chicken

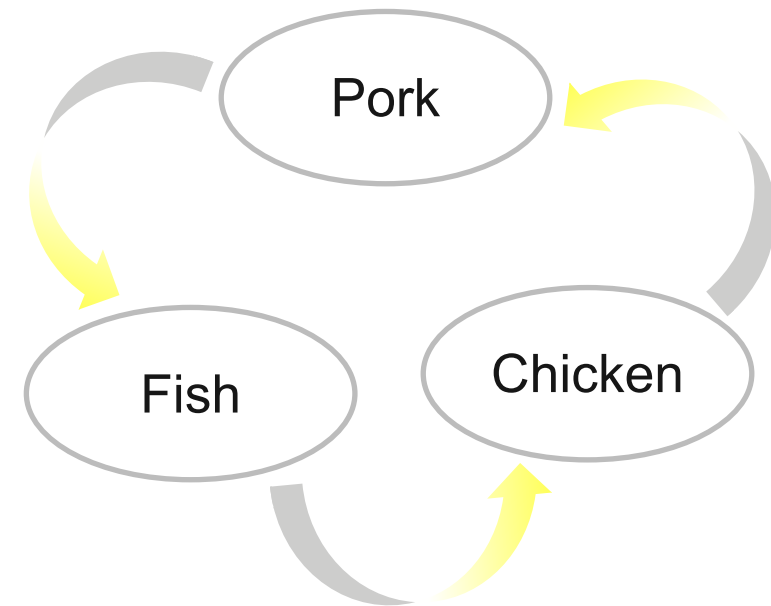
Can we do a **prediction** model?



Probabilistic modeling, an example

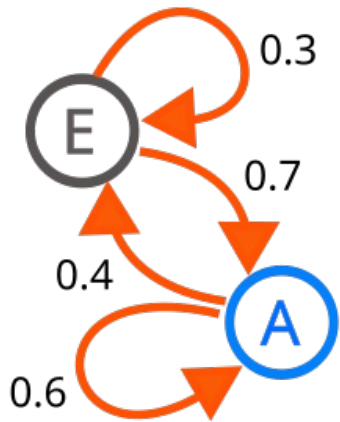
I noticed that Simple Servings at the U is doing a rotation of proteins: Pork, Fish and Chicken

Can we do a **prediction** model?



Probabilistic modeling, an example

Step 1: Markov chain



https://en.wikipedia.org/wiki/Markov_chain

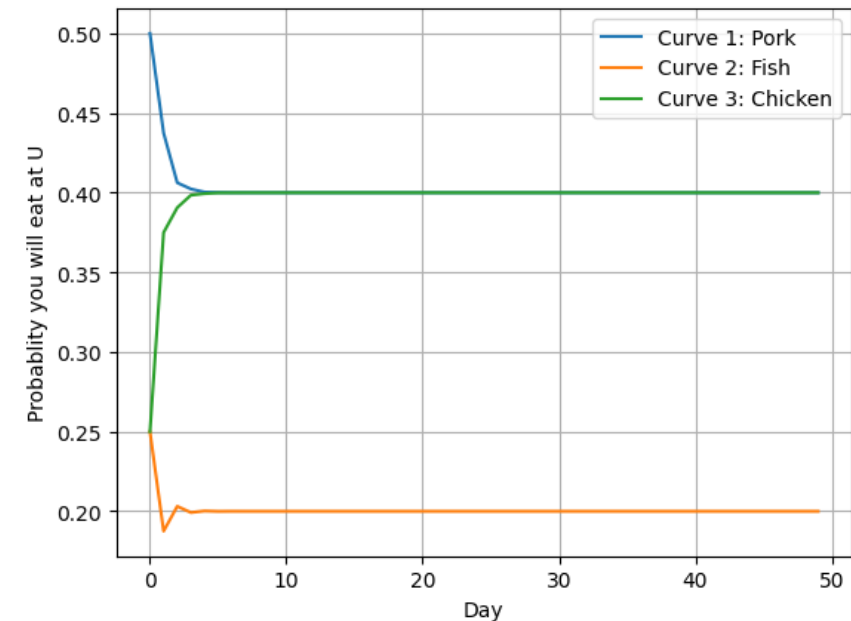
Step 2: Iteration in steps... in Python

```
A = np.array([[0.5, 0.5, 0.25], [0.25, 0, 0.25], [0.25, 0.5, 0.5]])
print (A)

[[0.5  0.5  0.25]
 [0.25 0.   0.25]
 [0.25 0.5  0.5 ]]

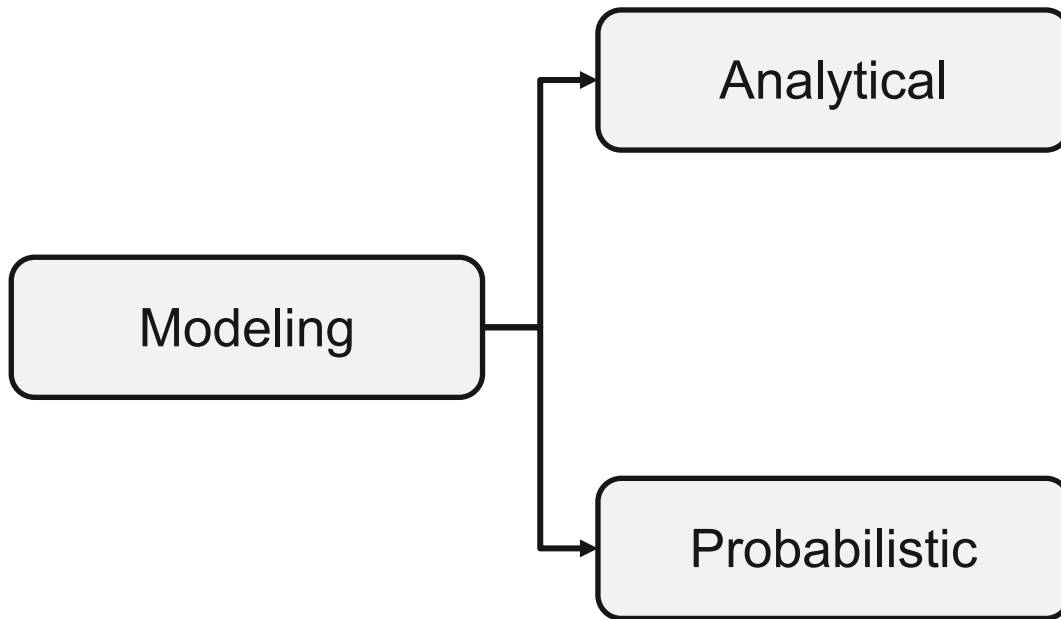
xtoday = [1, 0, 0]

for i in range(50):
    xtomorrow = A @ xtoday # xtomorrow is linear combination of xtoday
    xtoday = xtomorrow
    the_food[i, :] = xtomorrow
```



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Episode review and homework (?)



First Principle

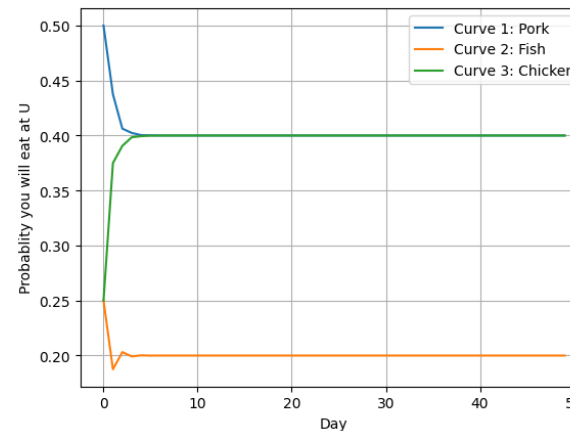
(1.10)

Power law

part of (4.2)

Chain rule

to get (1.47)



This is something off Taylor's textbook, but we will have Google Colab notebook to guide you through.



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